22. Answer : option(b)30min

$t_{75\%} = 2t_{50\%}$

$t_{50\%} = \left(\frac{t_{75\%}}{2}\right) = \left(\frac{60}{2}\right) = 30\,\text{min}$

23. Answer : option(d)

in140 days $\Rightarrow$ initial concentration reduced to $\left(\frac{1}{2}\right)g$

in 280 days $\Rightarrow$ initial concentration reduced to $\left(\frac{1}{4}\right)g$

in 420 days $\Rightarrow$ initial concentration reduced to $\left(\frac{1}{8}\right)g$

in 560 days $\Rightarrow$ initial concentration reduced to $\left(\frac{1}{16}\right)g$

24. Answer : option(b)

For a first order reaction

$t_{1/2} = \frac{0.6932}{k}$

For a second order reaction

$t_{1/2} = \frac{2^{n-1}-1}{(n-1)\,k\,[A_0]^{n-1}}$

$n = 2$

$t_{1/2} = \frac{2^{2-1}-1}{(2-1)\,k\,[A_0]^{2-1}}$

$t_{1/2} = \frac{1}{k\,[A_0]}$

25. Answer : option(c)

$1 \xrightarrow{t_{1/2}} \left(\frac{1}{2}\right) \xrightarrow{t_{1/2}} \left(\frac{1}{4}\right) \xrightarrow{t_{1/2}} \left(\frac{1}{8}\right) \xrightarrow{t_{1/2}} \left(\frac{1}{16}\right)$

$\therefore\ 4\,t_{1/2} = 2$ hours

$t_{1/2} = 30$ min

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